



CHAPTER 1: BASICS OF DERIVATIVES

NISM Series-VIII — Equity Derivatives | EasyOptionLearning

CHAPTER 1: BASICS OF DERIVATIVES

You will learn:

- What a derivative is and what an underlying asset means.
- Major derivative products: forwards, futures, options, and swaps.
- How the derivatives market evolved globally and in India.
- The roles of hedgers, speculators, and arbitrageurs.
- OTC vs exchange-traded derivatives and the key risks involved.

Basics of Derivatives

A **derivative** is a financial contract or product whose value is "derived" from the value of another asset. This base asset is known as the **underlying**. You can think of a derivative like a shadow; it simply moves based on the object (the underlying asset) casting it.

Underlying Asset: The asset from which a derivative derives its value. Examples include stocks, indices, commodities, currencies, and interest rates.

Derivatives can be based on a wide variety of underlying assets, including:

- **Metals:** Gold, Silver, Copper, Zinc, etc.
- **Energy:** Crude Oil, Natural Gas, Electricity.
- **Agri-commodities:** Wheat, Sugar, Coffee, Cotton.
- **Financial Assets:** Shares, Bonds, and Foreign Exchange (Currencies).

Derivatives Market – History & Evolution

While derivatives seem like a modern invention, their history dates back centuries. Here is a quick look at how they evolved:

Period	Key Development
12th Century	In European trade fairs, sellers signed contracts promising the future delivery of items.
13th Century	English Monasteries frequently sold their wool to foreign merchants up to 20 years in advance.
1634–1637	Tulip Mania in Holland; fortunes were lost after a speculative bubble in tulip futures popped.
Late 17th Century	A rice futures market was developed in Dojima, Japan, to protect farmers from adverse weather or warfare.
1848	The Chicago Board of Trade (CBOT) began facilitating the trading of forward contracts on various commodities.
1865	The CBOT listed the very first exchange-traded futures contracts.
1973	The Chicago Board of Options Exchange (CBOE) became the first marketplace for trading listed options.
1982	The first stock index futures were launched by the Kansas City Board of Trade.

Why did derivatives grow so massively worldwide?

- Increased fluctuations (volatility) in asset prices.
- Global integration of financial markets.
- Advanced technology reduces transaction costs.

- Better understanding of risk management tools.
- Continuous innovations in new products.

Indian Derivatives Market

In India, the journey of organised derivatives trading began in the late 1990s:

Year	Key Event / Development
1996	The SEBI set up the Dr. L.C. Gupta Committee to develop a regulatory framework.
1998	The Prof. J.R. Varma Committee recommended measures for risk containment.
1999	The Securities Contract Regulation Act (SCRA) was legally amended to classify "derivatives" as "securities".
2000	Exchange-traded derivatives officially started with Index Futures based on the Nifty and Sensex.
2001	Options on indices, options on individual stocks, and futures on individual stocks were subsequently launched.
2013	The Metropolitan Stock Exchange of India Limited (MSEI) also started offering equity derivatives in February 2013.

Products in the Derivatives Market

There are four primary building blocks in the derivatives world:

Derivative Type	Key Characteristics / Description
Forwards	A private, customised agreement (Over-the-Counter) between two parties to buy or sell an asset at a future date for a pre-decided price. Both parties are obliged to honour the transaction.

Derivative Type	Key Characteristics / Description
Futures	Standardised forward contracts are traded on a regulated exchange. Contract specifications such as lot size and expiry are fixed by the exchange.
Options	A contract that gives the buyer the right, but not the obligation, to buy or sell the underlying asset. The buyer pays a premium for this right, while the seller has an obligation if the right is exercised.
Swaps	A series of forward contracts is used to manage risks associated with volatile interest rates, currency exchange rates, and commodity prices.

Note: *Forwards and Swaps are generally OTC products, whereas Futures and Options are primarily exchange-traded products.*

Market Participants

The derivatives market is driven by three main types of players:

- **Hedgers:** These are participants who face price risks in the real world (like a farmer worried about falling crop prices) and use derivatives to protect themselves or reduce their risk.
- **Speculators/Traders:** They try to predict future price movements to make a profit. They love derivatives because they offer leverage and are generally cheaper and faster to trade than the actual underlying assets.
- **Arbitrageurs:** They hunt for price differences of the same product across two different markets. By buying cheap in one location and simultaneously selling high in another, they lock in a profit.

Types of Derivatives Market

Derivatives can be traded in two main ways: Over-The-Counter (OTC) and Exchange-Traded.

Feature	Over-the-Counter (OTC) Market	Exchange-Traded Market
Contract Type	Tailor-made / Customised to specific needs	Standardized
Trading Platform	Private networks or telephone	Anonymous auction on organised exchanges
Counterparty Risk	High (Decentralised risk management)	Low (Clearing corporation guarantees trades)
Regulation	Less regulated; mostly among qualified counterparties	Highly regulated for overall market safety
Transparency	Private, with little to no public disclosure	Public, with highly transparent pricing

Significance of Derivatives

Why do we even need this market? It serves several vital functions for the economy:

- **Price Discovery:** It helps the market discover fair prices based on actual valuations and expectations.
- **Risk Transfer:** It transfers various risks from those who want to avoid them (Hedgers) to those who have the appetite to take them (Speculators).
- **Market Stability:** It shifts speculative, high-risk trades from unorganised markets into a highly regulated and organised space.
- **Leverage:** Derivatives allow participants to take larger exposures with a relatively small initial investment (margin).

Various Risks Faced by Participants in Derivatives

Before jumping in, every beginner must know about a highly practical safeguard: the **Model Risk Disclosure Document**. Provided by your broker at the time of signing the agreement, this document officially outlines all the dangers of trading leveraged instruments. You must carefully read it before taking your first trade.

These are the primary risks outlined that traders face:

- **Counterparty Risk (Default/Credit Risk):** The risk of economic loss if the other party completely fails to fulfil their contractual obligation.
- **Price Risk:** The risk of financial loss because the asset's price moved against your position.
- **Liquidity Risk:** The inability to easily exit from a position due to a lack of buyers or sellers in the market.
- **Legal/Regulatory Risk:** The risk that a contract may not be legally enforceable.
- **Operational Risk:** Losses resulting from fraud, inadequate documentation, or improper execution.

Key Summary

- A **derivative** is a financial contract that gets its value from an underlying asset like stocks, metals, or agriculture.
- The **Chicago Board of Trade (CBOT)** holds significant historical importance; it facilitated forwards in 1848 and listed the very first exchange-traded futures in 1865.
- The four main derivative products are **Forwards, Futures, Options, and Swaps**.
- **Futures** have rigid limitations—they cannot be customised, have restricted maturities, and carry daily MTM administrative costs.
- The three main participants are **Hedgers** (avoid risk), **Speculators** (take risk for profit), and **Arbitrageurs** (seek price gaps).
- All beginners must read the **Model Risk Disclosure Document** to fully grasp leverage and market risks before trading.

Common Mistakes

- **Ignoring Leverage:** Beginners often trade too many lots because the margin is low, forgetting that derivatives are highly leveraged and a small price drop can cause massive losses.
- **Forgetting Contract Specifications:** Newbies might forget that they cannot customise a Futures contract. They must trade in standard lots (e.g., NIFTY Lot Size = 65) and must close positions before the strict Tuesday Expiry.
- **Confusing Options with Futures:** Thinking you must buy the asset in an Options contract. Remember: Options give the buyer the right to walk away, whereas Futures create a strict obligation to settle the trade.
- **Trading without reading the Risk Document:** Jumping into trades without understanding Counterparty Risk or Liquidity risk, assuming you can always sell whenever you want.

Practical Task: "Spot the Difference!"

1. Open a financial website (like NSE India) or your broker's app.
2. Search for the current NIFTY 50 index value (this is the actual Spot Price).
3. Next, search for a current NIFTY Futures contract.
4. Observe the Limitations: Check the contract details. Try to change the quantity to "1" or pick a random expiry day. You can't! Confirm that the contract is strictly standardised (e.g., NIFTY Lot Size = 65 and NIFTY Expiry = Tuesday).
5. Compare: Look at the exact price of the Futures contract versus the Spot Price of the index. Notice how the futures price trades slightly differently compared to the spot price — this gap is known as the "basis"!

Frequently Asked Questions (FAQ)

Q1. What is a derivative in simple words?

A derivative is a financial contract whose value depends on another asset, called the underlying. If the price of the underlying changes, the value of the derivative changes as well.

Q2. Why do derivatives exist?

Derivatives help participants manage risk, speculate on price movements, and discover fair market prices by allowing risk to be seamlessly transferred from one participant to another.

Q2. What is the difference between a hedger and a speculator?

A hedger uses derivatives to reduce risk, while a speculator uses derivatives to profit from expected price movements. One seeks protection; the other seeks opportunity.

Knowledge Check

Quiz 1: A derivative derives its value from:

- A) Government regulations
- B) An underlying asset
- C) Exchange membership
- D) Interest rates only

Answer: B

Explanation: Derivatives obtain their value from an underlying asset such as stocks, indices, commodities, or currencies.

Quiz 2: Who primarily uses derivatives to reduce risk?

- A) Speculator
- B) Arbitrageur

- C) Hedger
- D) Investor

Answer: C

Explanation: Hedgers use derivatives to protect themselves against adverse price movements.

Quiz 3: Which contract gives the buyer a right but not an obligation?

- A) Forward
- B) Future
- C) Swap
- D) Option

Answer: D

Explanation: Option buyers can choose whether or not to exercise their rights to walk away.

Chapter Complete! You have finished Chapter 1. Continue to Chapter 2: Understanding Futures Contracts to build on these foundations.